

Gene Pulser® Electroprotocols

* We recommend adapting this protocol to use the Gene Pulser electroporation buffer (catalog #165-2676, 165-2677), which increases cell viability and transfection efficiency in mammalian cell lines.

Cell Type	Mammalian, suspension	Molecules Electroporated	DNA
Species Used	Mouse, D10.G4.1, T-cell, helper		

Before the Pulse

Cell Growth Medium	DMEM (10% fetal calf serum) + 20% con A supplement (GIBCO/BRL, Sigma)	Growth Phase at Harvest	log phase
		Pre-pulse Incubation	10 to 15 min on ice
Wash Solution	Phosphate Buffered Saline without Ca++, Mg++		

The Pulse

Electroporation Temperature	Room Temperature		
Electroporation Medium*	Ca++, Mg++ free	Cuvette Gap	0.4 cm
Cell Density	10 (7) cells / 800 µl	Voltage	0.3 kV
Volume of Cells	800 µl	Field Strength	0.95 kV/cm
DNA Concentration	10 µg in sterile water or TE	Capacitor	960 µF
DNA Resuspension Buffer	Not given	Resistor	(Pulse Controller) Ω none
Volume of DNA	10 to 20 µl	Time Constant	11 msec
After the Pulse			
Outgrowth Medium	DMEM (10% serum) + 20% con A supplement		

Outgrowth Temperature	37 °C	Relevant Publications and/or Comments Note: exponential values designated in parentheses. PBS: 1x = 8g NaCl, 0.2g KCl, 0.2g KH ₂ PO ₄ , 1.15g Na ₂ HPO ₄ HBS: 10mM HEPES, pH 7.2, 150 mM NaCl, 5 mM CaCl ₂	
Length of Incubation	2 days		
Selection Method or Assay Used	Hygromycin		
Electroporation Efficiency	not quantified (used CAT assay)		
Per Cent Survival	70 to 80%		

Name of Submitter Yang Park, Postdoc

Institution Address UMDNJ-RWJMS
Pathology
675 Hoes Lane
Piscataway, NJ 08854

Survey Number
143